

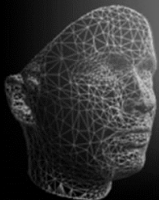
Advanced Topics in Computer Graphics II

Geometry Processing

Shape Correspondences

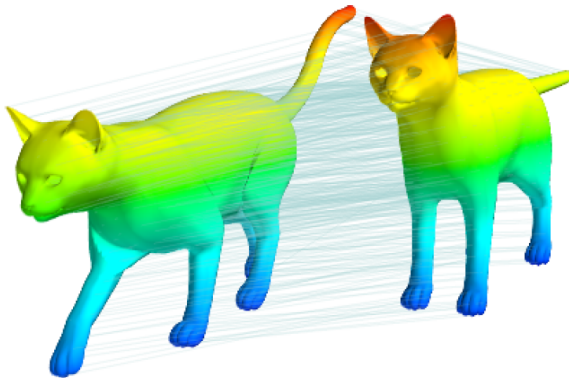


January 14, 2025



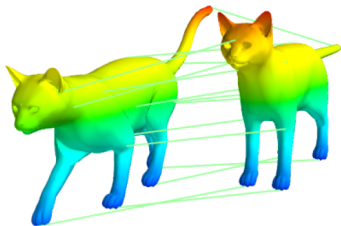
Problem

Given 3D surface meshes with or without corresponding feature points, find a dense and bijective mapping with as little distortion as possible.

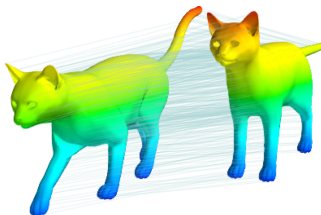




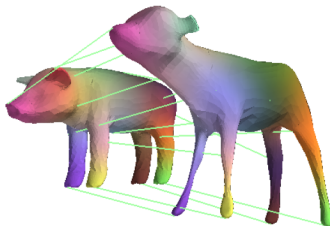
Correspondences



(a) sparse/isometric



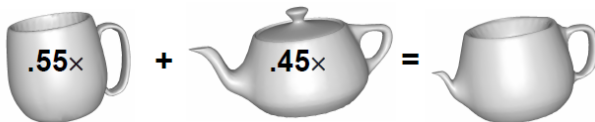
(b) dense/isometric



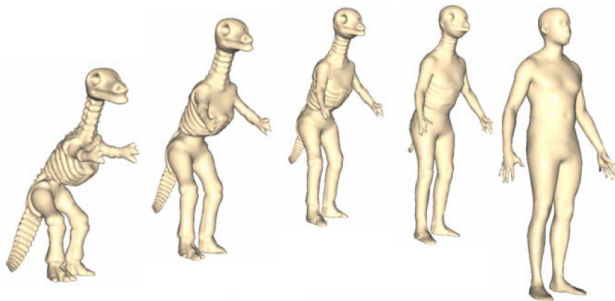
(c) sparse/near-isometric



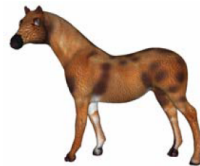
Applications



Morphing



Shape Space



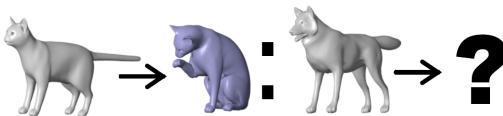
Texture Transfer



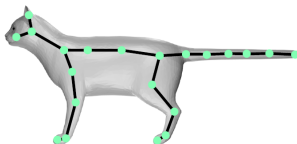
Applications



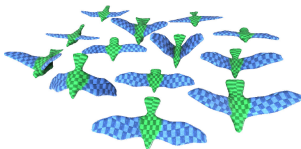
Interpolation



Deformation / Pose Transfer



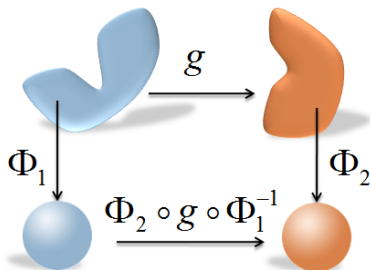
Skeleton Extraction



Sampling / Compression



- ▶ How can we achieve pose invariance?
 - ▶ Use intrinsic properties!
 - ▶ Möbius Voting¹
- ▶ Based on well-established ideas from conformal parameterization.
- ▶ Parameterize both surfaces over the same domain, then map the parameterizations.



¹Y. Lipman and T. Funkhouser. "Möbius voting for surface correspondence". In: *ACM Transactions on Graphics (TOG)* 28.3 (2009), pp. 1–12.

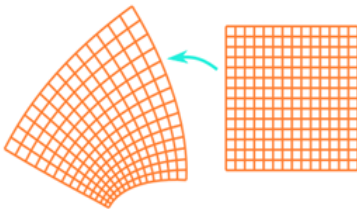


Intrinsic Properties

- ▶ Intrinsic properties: Based only on the first fundamental form.
- ▶ The properties are only based on local distance and angle.
- ▶ They are, for example, *not* based on mean curvature, principal curvature directions.
- ▶ Such properties are mostly preserved for pose changes
- ▶ Example: The properties are fully preserved for deformations of a sheet of paper.

Conformal Parameterization

- ▶ A conformal mapping preserves angles.
- ▶ When two curves are crossing each other, the angle between them must be the same after mapping them to the new domain.



A parameterization maps between two surfaces



A conformal parameterization of a 3D model

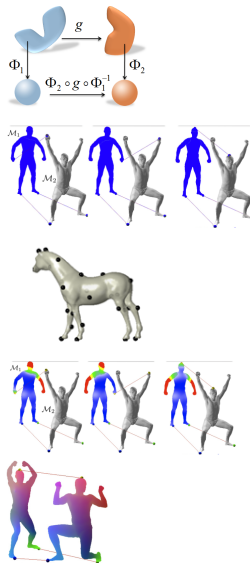
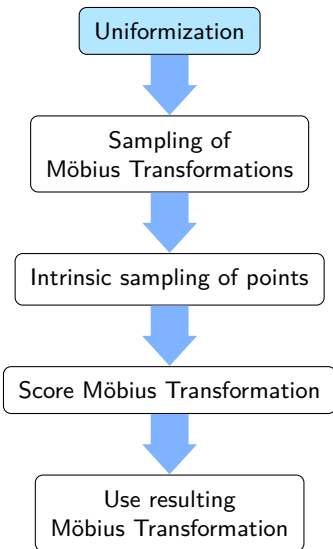
Why do we use conformal parameterizations?

- ▶ Conformal mappings are well understood.
- ▶ The space of conformal mappings between two genus-0 surfaces is small (6 Dimensions).
- ▶ The calculation of a conformal mapping is fast.
- ▶ Isometric mappings are a subset of conformal maps.





Algorithm Overview

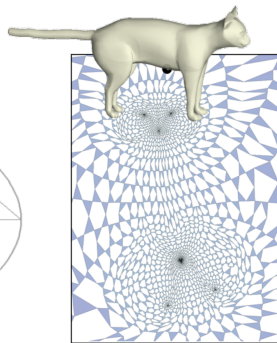
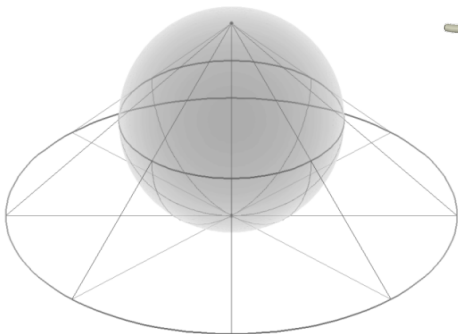




Uniformization

Each genus 0 surface can be mapped conformally to the extended complex plane.
– “The Uniformization Theorem” (Abikoff 1981)²

- ▶ The extended complex plane is $\bar{\mathbb{C}} = \mathbb{C} \cup \infty$.
- ▶ By a stereographic projection, the $\bar{\mathbb{C}}$ is conformal to a sphere.



²W. Abikoff. “The uniformization theorem”. In: *The American Mathematical Monthly* 88.8 (1981), pp. 574–592.

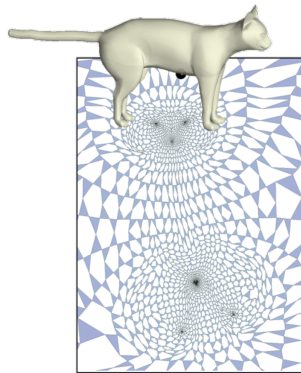


Definition: Möbius Transformation

$$\phi : \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}$$

$$z \mapsto \frac{az + b}{cz + d} \text{ with } ad - bc \neq 0$$

- ▶ ϕ is conformal
- ▶ ϕ is a combination of:
 - ▶ Scaling
 - ▶ Rotation
 - ▶ Translation
 - ▶ Inversion





Möbius Transformations

$$f(z) = \frac{az + b}{cz + d}$$



Definition: Möbius Transformation

$$\phi : \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}$$

$$z \mapsto \frac{az + b}{cz + d} \text{ with } ad - bc \neq 0$$

► Set of Möbius Transformations:

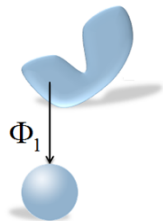
$$M : \{\phi : \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}} \mid \phi \text{ is a Möbius Transformation}\}$$

► Properties:

- $(M, \circ)$ is a group (closed, identity, inverse).
- M contains *exactly* the conformal mappings in $\overline{\mathbb{C}}$.
- M is a 6-dimensional $\mathbb{R}$ -vector space (normalization).

How many possibilities are there to map a genus-0 surface onto the extended complex plane conformally?

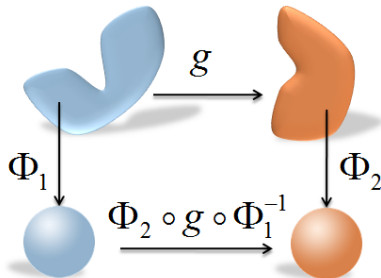
- Isomorphic to Möbius Transformations.
- *Proof:* Assume there are two different Uniformizations:
 $u_1, u_2 : S \rightarrow \overline{\mathbb{C}}$. Then $u_1 \circ u_2^{-1} \in M$.
- After factoring out, the Möbius Transformation Uniformization is unique.



Question

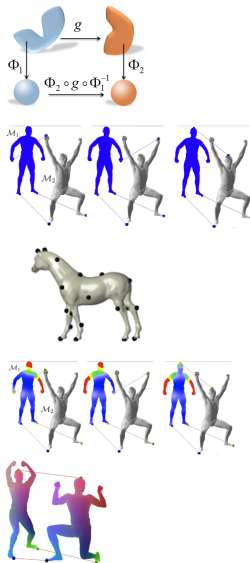
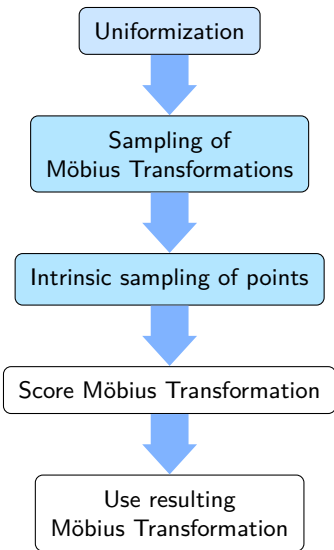
How many possibilities are there to conformally map a genus-0 surface onto another?

- ▶ M is a 6-dimensional $\mathbb{R}$ vector space.
- ▶ Fix the mapping of 3 points from one surface onto the other.
- ▶ Then there is exactly *one* conformal map possible.





Algorithm Overview





- ▶ Möbius Transformations are defined in a 6D space.
- ▶ 3 points and their maps are required.

Idea

- ▶ Select a small set of points based on intrinsic properties.
- ▶ Try all/some possible permutations.

Possibilities:

- ▶ Maxima of the Gaussian
- ▶ Minima of the Gaussian
- ▶ Maxima of the Average Geodesic Distance (AGD)

Definition: Average Geodesic Distance

$$\text{AGD}(x) = \frac{\int dS \, d(x, x')}{\int dS}$$

- ▶ Map each point to the average geodesic distance point on the other mesh patches.
- ▶ Take maxima of AGD as sample points: $\text{argmax}_x \text{AGD}(X)$
- ▶ The result is only based on intrinsic properties.

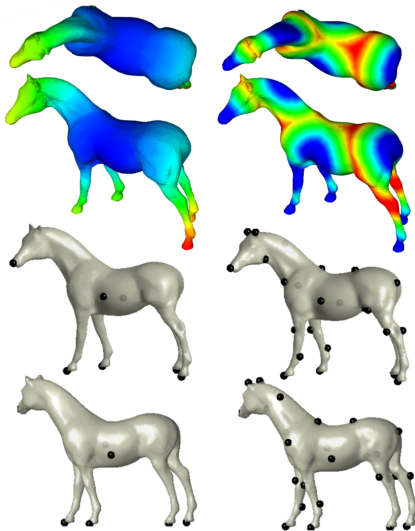


Definition: Minimal Geodesic Distance

The minimal Geodesic Distance for a point set S_1 is

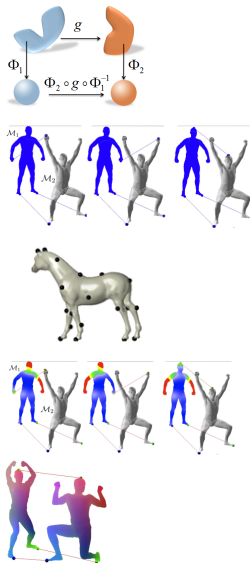
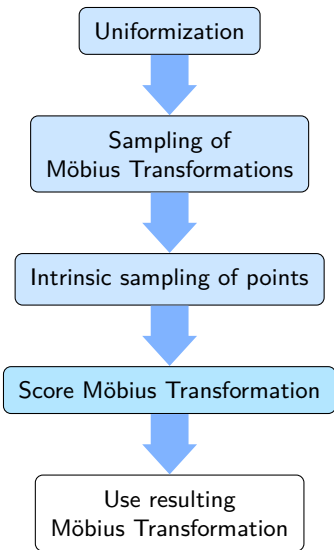
$$\text{MGD}(x, S_1) = \min_{q \in S_1} d(x, q)$$

- The refined point set S_2 contains the minima and maxima of MDG.



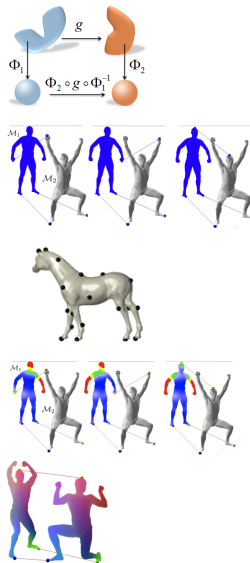
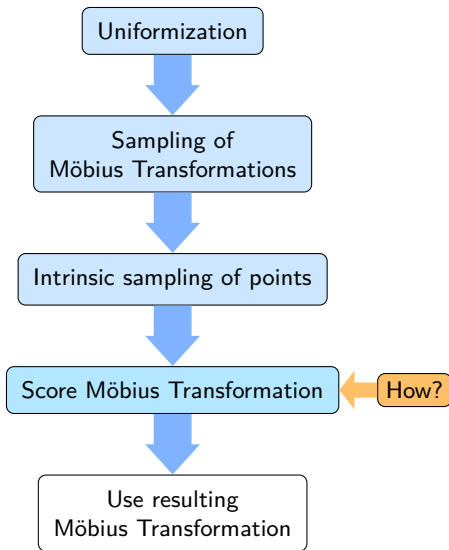


Algorithm Overview





Algorithm Overview

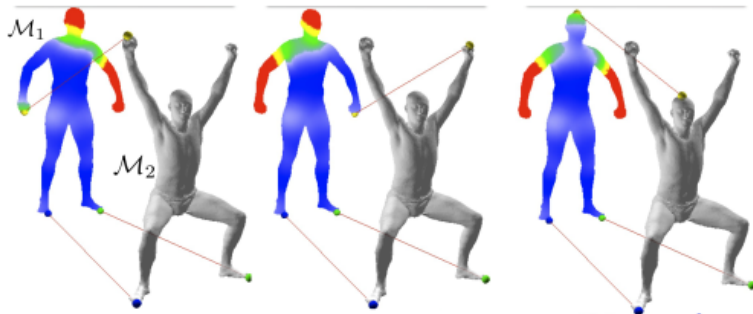




How to find good maps?



- ▶ In the end, we look for near isometric maps.
- ▶ We measure isometry of the map.
- ▶ Then we compare geodesic distances of the feature points of the original and the target shape.
- ▶ What is the best map depends on the application.

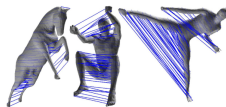




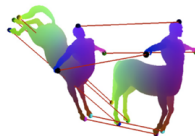
Correspondence Finding



Symmetry Detection



Surface Mapping





Lipman, Funkhouser (2009)³:

1. Randomly draw a map.
2. Find mutual closest pairs c of the feature points.
3. Compute the Error: $E(c) = \sum_k \|z_k - c(z_k)\|$
4. Update correspondence matrix: $C_{k,l} = C_{k,l} + \frac{1}{\epsilon + E(c)/\|c\|}$
5. Greedy look for best matches and remove the corresponding column and row.

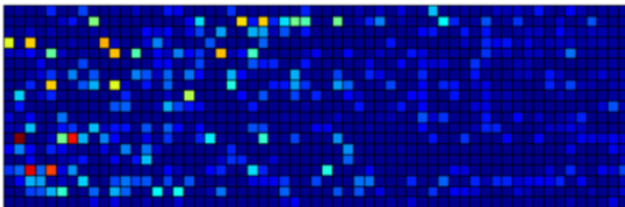


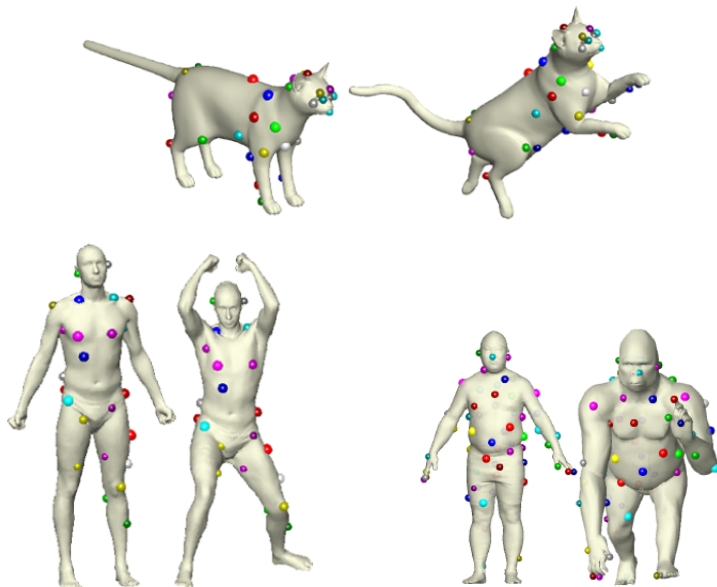
Figure: The correspondence matrix

Remark: The error $E(c)$ is the distance on the Riemann sphere.

³Y. Lipman and T. Funkhouser. "Möbius voting for surface correspondence". In: *ACM Transactions on Graphics (TOG)* 28.3 (2009), pp. 1–12.



Correspondence finding

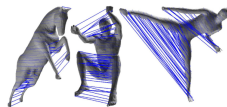




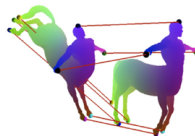
Correspondence Finding



Symmetry Detection



Surface Mapping





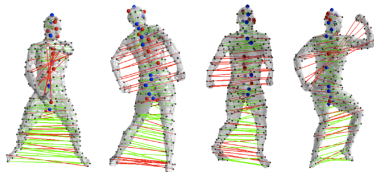
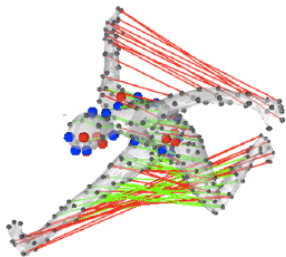
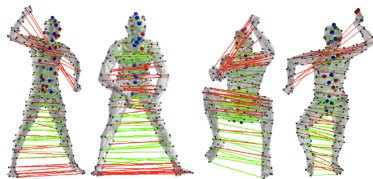
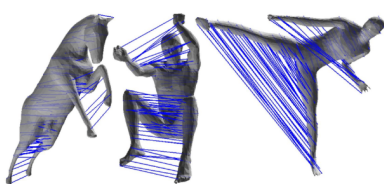
Kim et al. (2010)⁴:

- ▶ Generate Maps using feature points $p \in F$ with $|F| \approx 15$.
 - ▶ For all tuples $(p_1, p_2, p_3) \in F^3$, generate $p_1 \rightarrow p_1, p_2 \leftrightarrow p_3$
 - ▶ For all tuples $(p_1, p_2, p_3, p_4) \in F^4$, generate $p_1 \rightarrow p_1, p_2 \rightarrow p_3 \rightarrow p_4$, and $p_1 \rightarrow p_2 \rightarrow p_3 \rightarrow p_4$
- ▶ Prune maps:
 - ▶ For matching points, AGD should be similar.
 - ▶ p, q are initially matching points ($q = m(p)$) with for a map m .
 - ▶ We require $\gamma < \frac{\text{AGD}(p)}{\text{AGD}(q)} < \frac{1}{\gamma}$ with $\gamma \approx 0.9$.
- ▶ For all points, the geodesic distances should be similar.
 - ▶ We require $\delta < \frac{d(p,q)}{d(m(p),m(q))} < \frac{1}{\delta}$ with $\delta \approx 0.9$
- ▶ Score maps:
 - ▶ Factors: Dense, isotropic, iso-distance surface sampling.
 - ▶ Map sampling: Count number of mutual closest pairs (represent the surface area mutual closest).

⁴V. G. Kim, Y. Lipman, X. Chen, et al. "Möbius transformations for global intrinsic symmetry analysis". In: *Computer Graphics Forum*. Vol. 29. 5. Wiley Online Library. 2010, pp. 1689–1700.



Symmetry Detection

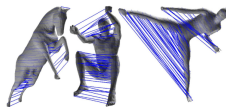




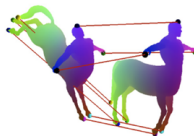
Correspondence Finding



Symmetry Detection



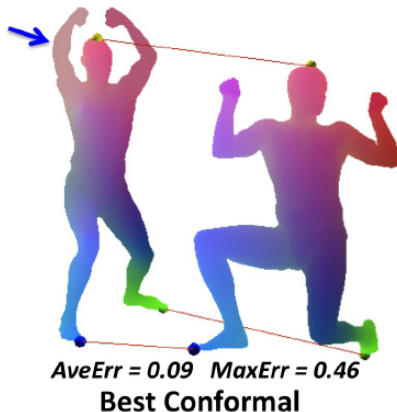
Surface Mapping





Blended intrinsic maps (Kim et al., 2011)⁵

- ▶ Use conformal maps to find correspondences.
- ▶ We could just take the best map.
- ▶ Can we do better?



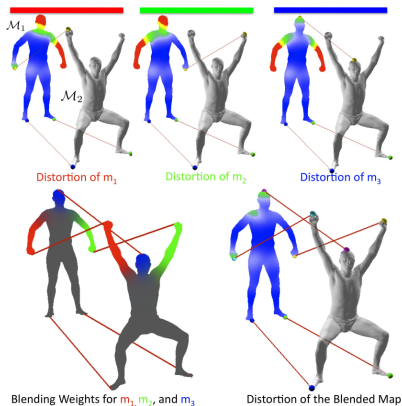
⁵V. G. Kim, Y. Lipman, and T. Funkhouser. "Blended intrinsic maps". In: *ACM transactions on graphics (TOG)* 30.4 (2011), pp. 1–12.

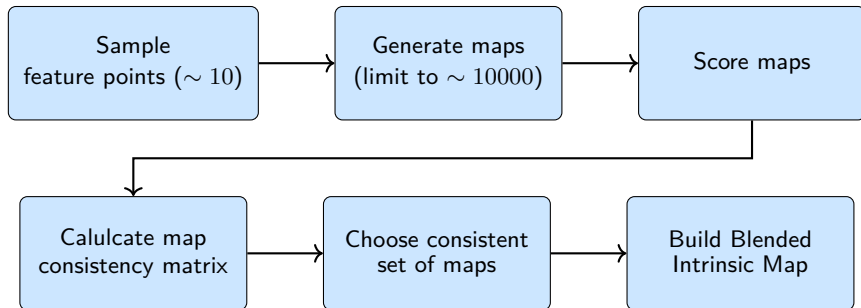


Surface mapping



- Observation: A map is locally mostly describing a near-isometric match.
- Idea: Blend between best local maps





To score a map m , generate 256 evenly spread points $p \in P$ and equal sized neighborhoods N_p .

$$c_m(p) = 2 / \left(\frac{\text{area}(N_p)}{\text{area}(m(N_p))} + \frac{\text{area}(m(N_p))}{\text{area}(N_p)} \right)$$

It holds $c_m(p) \leq 1$ and $c_m(p) = 1 \Leftrightarrow \text{area}(N_p) = \text{area}(m(N_p))$, i.e., for $c_m(p) = 1$, the area of the neighborhood N_p and the mapped neighborhood $m(N_p)$ are matching.

The consistency at a point $p \in P$ is calculated as

$$\check{S}(p) = \exp \left(- \frac{d(m_j^{-1}(m_i(p)), p)}{\sigma^2} \right) \text{ with } \sigma \approx 0.5$$

The global consistency of the map $S_{i,j}$ is calculated by

$$S_{i,j} = \sum_{p \in P} c_i(p) c_j(p) \check{S}_{i,j}(p)$$



Choose a consistent set:

- Find weights w_i for the map m_i :
$$\operatorname{argmin}_{(w_i)_{i \geq 0}} \sum_i \sum_j w_i w_j S_{i,j} = \operatorname{argmin}_w (w^T S w)$$
- Solution: Find the eigenvector with the largest eigenvalue.

Blending weights: $b_i(p) = c_i(p)w_i$

Blended Mapping:

$$f(p) = \operatorname{argmin}_{p' \in \mathcal{M}_2} \left(\sum_{i=1}^K b_i(p) d_{\mathcal{M}_2}(p', m_i(p))^2 \right)$$



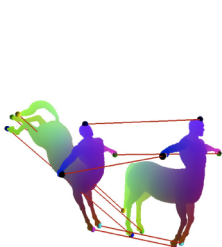
Blending Weights



Weighted Centroid



Surface Mapping



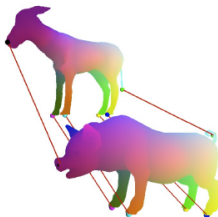
AveErr = 0.02 *MaxErr* = 0.09
Near-Isometric: TOSCA



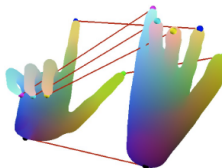
AveErr = 0.04 *MaxErr* = 0.16
Near-Isometric: SCAPE



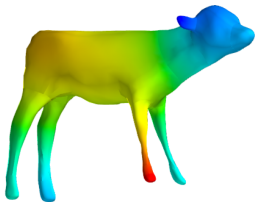
AveErr = 0.06 *MaxErr* = 0.19
Non-Isometric: Humans



AveErr = 0.05 *MaxErr* = 0.18
Non-Isometric: Animals



AveErr = 0.07 *MaxErr* = 0.19
Per Class: Hand



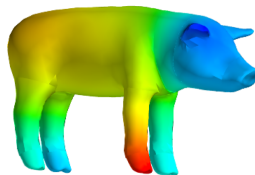
M

$$T : M \rightarrow N$$

→ point-wise map →

$$T_F : \mathcal{F}(M, \mathbb{R}) \rightarrow \mathcal{F}(N, \mathbb{R})$$

→ functional map →



N

$$T_F : \mathcal{F}(M, \mathbb{R}) \rightarrow \mathcal{F}(N, \mathbb{R}), \quad T_F(f) := f \circ T^{-1}$$

M. Ovsjanikov et al. "Functional maps: a flexible representation of maps between shapes". In: *ACM Transactions on Graphics (TOG)* 31.4 (2012), pp. 1–11



Definition: Functional Maps

Let M, N be two shapes, $T : M \rightarrow N$ a bijective mapping and $f : M \rightarrow \mathbb{R}$ a scalar function. Then the mapping

$$T_F : \mathcal{F}(N, \mathbb{R}) \rightarrow \mathcal{F}(M, \mathbb{R})$$
$$T_F(f) := f \circ T^{-1}$$

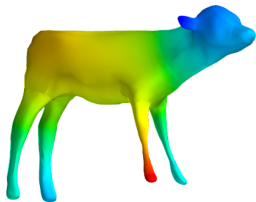
is called functional representation of T .



M. Ovsjanikov et al. "Functional maps: a flexible representation of maps between shapes". In: *ACM Transactions on Graphics (TOG)* 31.4 (2012), pp. 1–11



Pullback Functional Maps



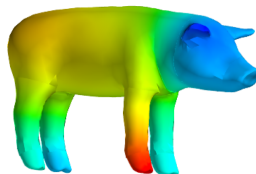
M

$$T : M \rightarrow N$$

→ point-wise map →

$$F_T : \mathcal{L}^2(N, \mathbb{R}) \rightarrow \mathcal{L}^2(M, \mathbb{R})$$

← pullback functional map ←



N

$$F_T : \mathcal{L}^2(N, \mathbb{R}) \rightarrow \mathcal{L}^2(M, \mathbb{R}), T_F(f) := f \circ T$$

Remark

For any fixed bijective map $T : M \rightarrow N$,

$T_F : \mathcal{F}(M, \mathbb{R}) \rightarrow \mathcal{F}(N, \mathbb{R})$ is a linear map between function spaces:

$$T_F(\alpha_1 f_1 + \alpha_2 f_2) = \alpha_1 f_1 \circ T^{-1} + \alpha_2 f_2 \circ T^{-1} = \alpha_1 T_F f_1 + \alpha_2 T_F f_2.$$

M. Ovsjanikov et al. “Functional maps: a flexible representation of maps between shapes”. In: *ACM Transactions on Graphics (TOG)* 31.4 (2012), pp. 1–11

O. Burghard et al. “Embedding shapes with Green’s functions for global shape matching”. In: *Computers & Graphics* 68 (2017), pp. 1–10



Functional Map Representation

Let $\{\phi_i^M\}$ and $\{\phi_i^N\}$ be a set of basis functions of $\mathcal{F}(M, \mathbb{R})$ and $\mathcal{F}(N, \mathbb{R})$, respectively. Then we can write any function $f \in \mathcal{F}(M, \mathbb{R})$ as $f = \sum_{i=1}^{\infty} a_i \phi_i^M$ and $T_F(f)$ as

$$\begin{aligned} T_F f &= \sum_{i=1}^{\infty} a_i \underbrace{T_F(\phi_i^M)}_{\in \mathcal{F}(N, \mathbb{R})} = \sum_{i=1}^{\infty} a_i \sum_{j=1}^{\infty} c_{ij} \phi_j^N \\ &= \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_i c_{ij} \phi_j^N \end{aligned}$$

In particular, c_{ij} is the j -th coefficient of $T_F(\phi_i^M)$ in the basis $\{\phi_i^N\}$. If the basis functions $\{\phi_i^N\}$ are orthonormal with respect to some inner product $\langle \cdot, \cdot \rangle$, we have $c_{ij} = \langle T_F \phi_i^M, \phi_j^N \rangle$.



Matrix Representation of T_F

Let $\{\phi_i^M\}$ and $\{\phi_i^N\}$ be a set of basis functions of $\mathcal{F}(M, \mathbb{R})$ and $\mathcal{F}(N, \mathbb{R})$, respectively and let $\{\phi_i^N\}$ be orthonormal with respect to an inner product $\langle \cdot, \cdot \rangle$. Then T_F is linear and can be represented as an (infinite) matrix $\mathbf{C}$

$$C_{ij} = \langle T_F \phi_i^M, \phi_j^N \rangle.$$

We write $T_F(f) = \mathbf{C}\mathbf{a}$ where $\mathbf{a} = (a_1, a_2, \dots)$.

Definition

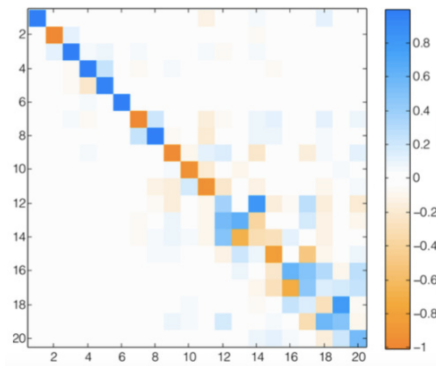
Let $\{\phi_i^M\}$ and $\{\phi_i^N\}$ be a set of basis functions of $\mathcal{F}(M, \mathbb{R})$ and $\mathcal{F}(N, \mathbb{R})$, respectively. A generalized linear functional mapping $T_F : \mathcal{F}(M, \mathbb{R}) \rightarrow \mathcal{F}(N, \mathbb{R})$ with respect to these bases is the operator defined by

$$T_F \left(\sum_{i=1}^{\infty} a_i \phi_i^M \right) = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_i c_{ij} \phi_j^N$$

where the c_{ij} define a possibly infinite matrix of real coefficients.



Matrix Representation



► Summary

- Let T_F be a linear map between functions.
- T_F has a compact representation as “sparse” matrix C .
- Least-squares problem: $C = \operatorname{argmin}_c \|Ca - b\|_2^2$

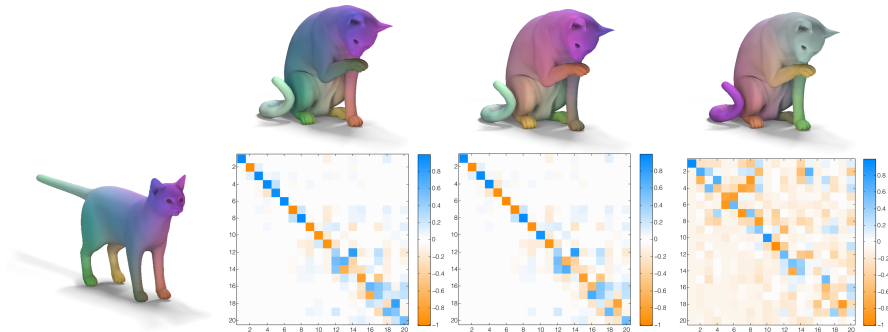


Linear functional mappings



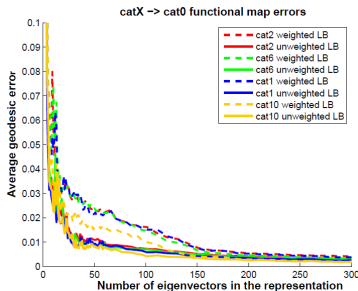
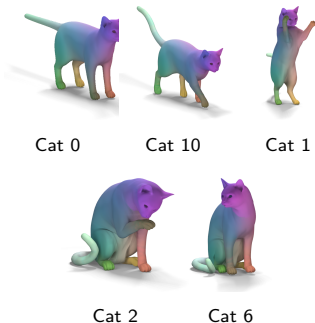
Example

Two shapes with three maps between them, each rendered as a point-to-point mapping through color correspondence (top) and its functional representation (bottom) with colors proportional to matrix values. Note that the least isometric map on the right-hand side leads to a less sparse functional matrix.





- Characteristics for choosing a basis
 - Compactness: The most natural functions on a shape should be well-approximated using a small number of basis elements.
 - Stability: The space of functions spanned by all linear combinations of basis functions must be stable under small shape deformations.
- For near isometric transformations, the first n ($n \approx 100$) Laplace-Beltrami eigenfunctions are a good choice as the basis for their functional representation.





- ▶ Laplace-Beltrami eigenfunctions as basis.
 - ▶ For an isometry T , the matrix elements c_{ij} are non-zero only if ϕ_i^M and ϕ_i^N correspond to the same eigenvalue.
 - ▶ In practice, we observe that if T is only approximately an isometry, the matrix $\mathbf{C}$ is still close to being sparse or funnel-shaped.

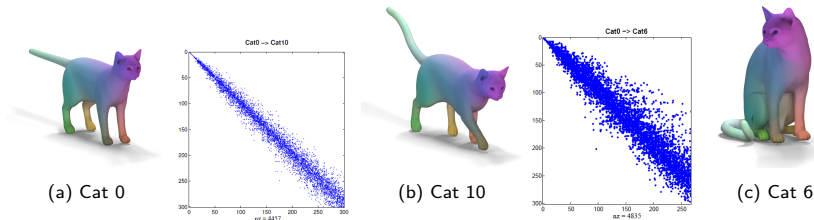


Figure: The sparsity pattern of the matrices $\mathbf{C}$ corresponding to two of the 4 maps shown before. Only cells where $C_{ij} > 0.11$ are shown. Note that more than 94% are not. Note also that the functional matrix for the more deformed shape “cat 6” is farther from being diagonal.



Function preservation

Let a and b be the representations of a pair of functions $f \in \mathcal{F}(M, \mathbb{R})$ and $g \in \mathcal{F}(N, \mathbb{R})$ with respect to the chosen bases of $\mathcal{F}(M, \mathbb{R})$ and $\mathcal{F}(N, \mathbb{R})$. The correspondence between f and g can be written as $\mathbf{C}a = b$, where $\mathbf{C}$ is the functional representation of the map.

- Note that the function preservation can be phrased entirely in terms of the matrix $\mathbf{C}$ without knowledge of the underlying correspondence T , since a and b do not depend on the map $\mathbf{C}$.
- If we can find enough function correspondences, we can recover $\mathbf{C}$.



- ▶ **Landmark point correspondence:** Given $T(x) = y$ with $x \in M, y \in N$
 - ▶ Construct indicator functions around x and y and distance functions f and g for the distance from x and y in M respectively N .
 - ▶ Set $Ca = b$.
- ▶ **Segment correspondences:** Similar to above using indicator or distance functions.
- ▶ **Utilization of additional information about the shapes:** A given symmetry $S : M \rightarrow M$ induces a symmetry operator $S_{\mathcal{F}} : \mathcal{F}(M, \mathbb{R}) \rightarrow \mathcal{F}(N, \mathbb{R})$ by $S_{\mathcal{F}}(f)(x) = f(S^{-1}(x))$.
- ▶ The symmetry operator, just like the Laplace-Beltrami operator or derived operators (e.g., the heat operator), should be preserved under isometries.
- ▶ Therefore, given two functional operators $S_{\mathcal{F}}, R_{\mathcal{F}}$ on M and N , respectively, we expect $R_{\mathcal{F}}C = CS_{\mathcal{F}}$. This gives an additional linear constraint.
- ▶ Every bijective map T has a functional representation $T_{\mathcal{F}} = C$. The opposite is not always true.
- ▶ In a natural setting, we expect orthonormal or nearly-orthonormal functional map matrices for a point-to-point map.
- ▶ In this case, we receive the inverse of C by transposition.



Theorem

If the basis functions are discrete and orthonormal or if the underlying map T is volume preserving, i.e., $\mu^M(x) = \mu^N(T(x))$, where μ^M and μ^N are volume elements on M and N respectively, then the matrix $\mathbf{C}$ must be orthonormal.

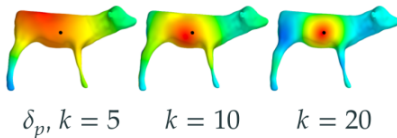


- ▶ Given two shapes M and N . Find the underlying bijective map T .
- ▶ **Approach:** First recover C and then calculate T .
- ▶ **Algorithm** for calculating C :
 - ▶ Construct a system of constraints, e.g., by matching landmark points.
 - ▶ Optimize $\min_C \|Ca - b\|$ with $a = (a_1, a_2, \dots, a_n)$ and $b = (b_1, b_2, \dots, b_n)$ vectors, such that $Ca_i = b_i$ is a linear constraint $\forall i \in \{1, \dots, n\}$.
- ▶ Now we want to get the underlying bijection T out of the functional representation C .
- ▶ **Approach 1 (indicator functions):**
 - ▶ For every $x \in M$ construct the indicator function $f : M \rightarrow \mathbb{R}$.
 - ▶ Calculate $g = T_F(f) = Ca$.
 - ▶ Declare $y \in N$ as the point where $g(y)$ obtains the maximum.
 - ▶ The approach requires $\mathcal{O}(V_M V_N)$ operators for meshes with V_M and V_N vertices.



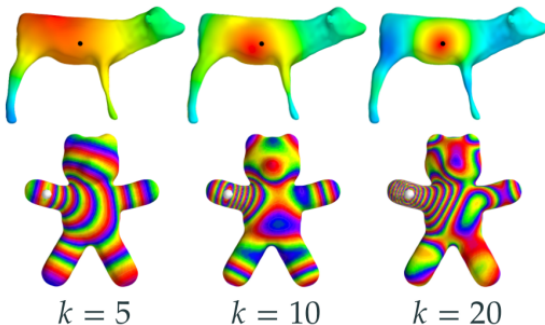
► Approach 2: Alignment of Delta-distributions

- Consider the delta function around x :
$$\delta_x = \lim_{t \rightarrow 0} k_t^M(x, \cdot) = \lim_{t \rightarrow 0} \sum_{i=0}^{\infty} e^{-t\lambda_i} \phi_i^M(x) \phi_i^M(\cdot).$$
- The coefficients of δ_x are $a_i = \phi_i^M(x)$.
- Write these coefficients as columns in a matrix ϕ^M for every $x \in M$. Doing the same for every $y \in N$ gives ϕ^N .
- $C\phi^M$ is the image of all delta functions in M .
- Finding the nearest neighbor of every point in $C\phi^M$ in ϕ^N leads us to a point-to-point correspondence.
- For each column in $x \in C\phi^M$ find the closest $\tilde{x} \in \phi^N$.
- Using a kd-tree structure, this requires only $\mathcal{O}(V_N \log V_N + V_M \log V_M)$ operations.



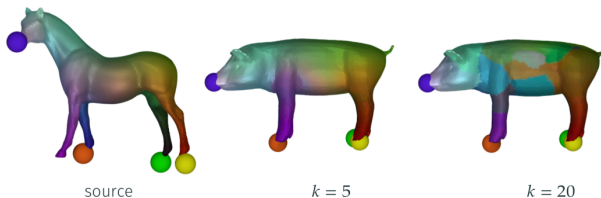
- Approximating δ -distributions (k eigenvectors).
- Point-wise functional map aligns δ -distributions: $T : M \rightarrow N \Rightarrow T_F \delta_p^M = \delta_{T(p)}^N$

Alignment of Delta-Distributions



- Distances are *dependent* on dimensions k (not well-defined).
- Distances become meaningless with increasing k .

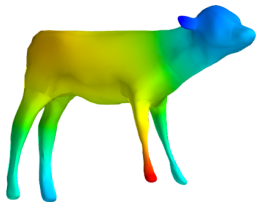
Alignment depends on Distances



- ▶ Results when using 4 sparse point constraints and alignment.
- ▶ $k = 5$: A L_2 -distance field approximates the intrinsic distances.
- ▶ $k = 20$: The L_2 -distance field becomes meaningless.



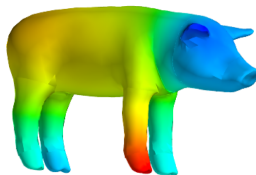
Non-Isometric shapes



M

$$T : M \rightarrow N$$

→ point-wise map →



N

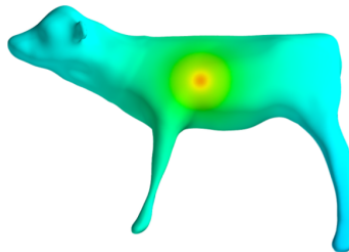
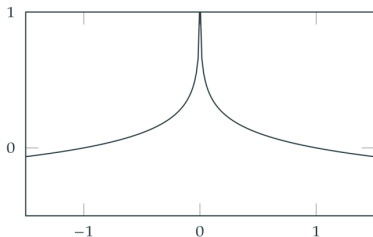
- ▶ Few functional constraints: $T_F(f) = ?$
- ▶ We want to keep the *linear behavior*.
- ▶ We want the distances to be *independent* of the dimension k .



Definition: Green's function

The Green's function g_p^M of Δ_M at a point p on the shape M is the solution of the differential equation

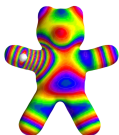
$$\Delta_M g_p^M = \delta_p^M$$



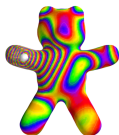
► Deltas:



$k = 5$



$k = 10$



$k = 20$

► Greens:



$k = 5$



$k = 10$



$k = 20$

Bottom: L_2 -distances $\left| g_p^M - g_q^M \right|_2^2$ approximate the Biharmonic distances.

Y. Lipman, R. M. Rustamov, et al. "Biharmonic distance". In: *ACM Transactions on Graphics (TOG)* 29.3 (2010), pp. 1–11

- The alignment G is an affine map related to the functional map T_F :

$$\begin{aligned} T_G(f) &= T_B(f) + t \\ &= \Delta_N^+ T_F^t \Delta_M(f) + \frac{1}{\sqrt{A_M}} \Delta_N^+ T_F^t \phi_1^M \end{aligned}$$

where Δ_N^+ is the pseudo-inverse of Δ_N .

- Linear constraints in T_F are linear constraints in T_G .
- The alignment can be determined by solving a least-squares problem.

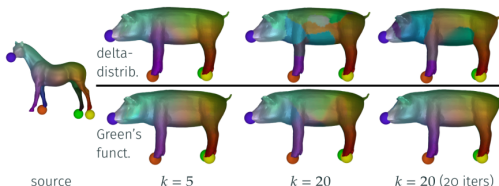
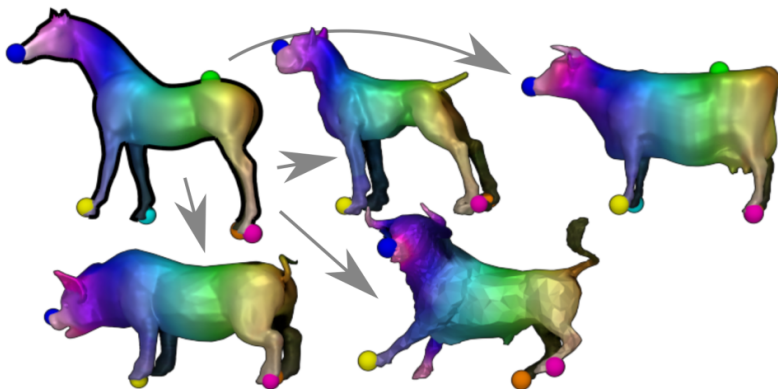


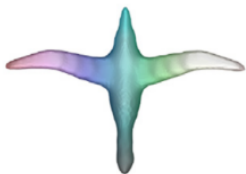
Figure: Top row: without and with alignment using ICP on δ_{ps} . Bottom row: Without and with alignment using ICP on g_{ps} .

O. Burghard et al. "Embedding shapes with Green's functions for global shape matching". In: *Computers & Graphics* 68 (2017), pp. 1–10

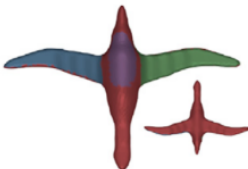


(Initial sparse point constraints are shown as colored balls)

O. Burghard et al. "Embedding shapes with Green's functions for global shape matching". In: *Computers & Graphics* 68 (2017), pp. 1–10



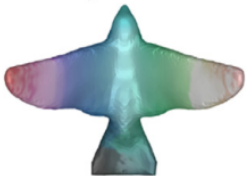
a) source



b) labels src



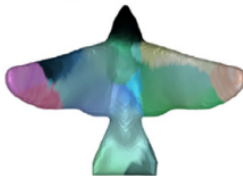
c) labels dst



d) our method



e) BIM



f) Fct.Maps

- ▶ Initial function constraints are shown as colored regions.
- ▶ BIM = Blendet intrinsic maps.



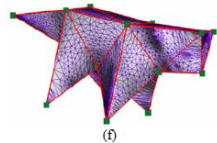
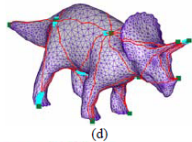
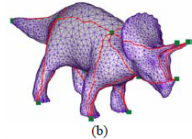
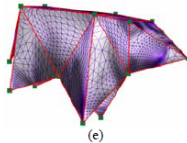
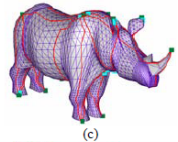
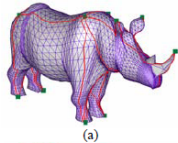
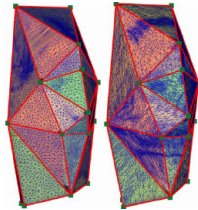
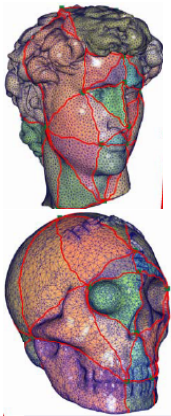
- **Non-Isometric case:** Alignment of Green's functions extend the functional maps.
- **(Near-)Isometric case:** Alignment of Green's functions requires fewer constraints.



Figure: Mesh Interpolation: Green's functions + Coherent Point Drift (CPD)

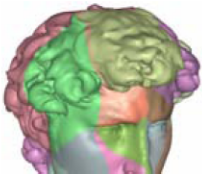
A. Myronenko and X. Song. "Point set registration: Coherent point drift". In: *IEEE transactions on pattern analysis and machine intelligence* 32.12 (2010), pp. 2262–2275

- First construct a base mesh from triangular patches.
- Then map the patches on base triangles.

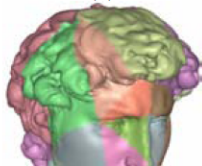


V. Kraevoy and A. Sheffer. "Cross-parameterization and compatible remeshing of 3D models". In: *ACM Transactions on Graphics (TOG)* 23.3 (2004), pp. 861–869

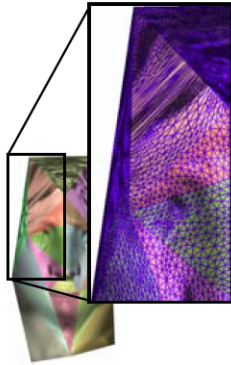
- Refine initial mapping by smoothing a parameterization across patch boundaries.



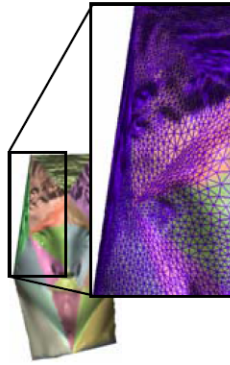
(a)



(b)



(c)



(d)



- A common base domain is constructed by incrementally adding pairs of matching edge paths between features.

Algorithm *PathMatch*

```

 $M_s' = M_s$ 
 $M_t' = M_t$ 
Compute the shortest paths  $s^i$  for each pair of vertices in  $V_s$ 
Compute the shortest paths  $t^j$  for each pair of vertices in  $V_t$ 
 $ST = \emptyset$ 
foreach  $s^i$ 
   $ST \leftarrow \langle s^i, t^j \rangle$  /* pairs of matching paths */
while  $ST \neq \emptyset$ 
   $\langle s, t \rangle = ST.RemoveShortest()$ 
  if NonBlocking( $s, t$ )
    Add  $s$  to  $P_s$ ; Add  $t$  to  $P_t$ 
    Remove all interior vertices of  $s$  from  $M_s'$ 
    Remove all interior vertices of  $t$  from  $M_t'$ 
    Update( $ST, s, t$ )
  end
end
end

```

Check for consistency:

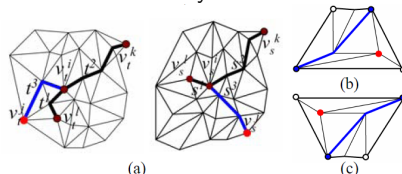
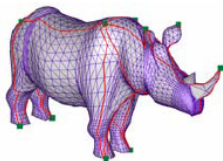


Figure 2: Path validity conditions. (a) Matching paths s^3 and t^3 (blue) violate cyclical order. Blocking paths (blue): (b) path in P_s ; (c) matching path in P_t . This pair of paths places the red feature vertex on different patches on P_s and P_t .

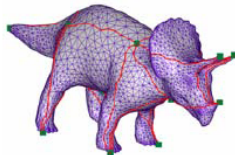
- Constructed patches are so far not necessarily triangular.
- Make each patch triangular by connecting different boundary loops (may require insertion of additional vertices) and triangulating the resulting simple patches.



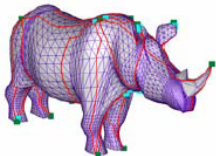
Common Base Domain



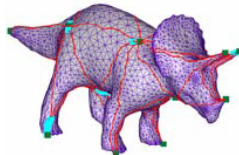
(a)



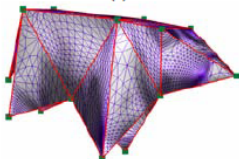
(b)



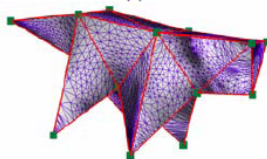
(c)



(d)



(e)



(f)

- ▶ Map each patch to the corresponding base triangle using mean value parameterization.
- ▶ Map each triangle of the source base mesh to the corresponding target triangle using barycentric coordinates.
- ▶ This defines an initial bijective mapping, which may still have a large distortion.
- ▶ For each vertex of the mesh:
 - ▶ Map the triangle and its one-ring onto the base domain.
 - ▶ Map the base domain onto an equilateral triangle.
 - ▶ Find the new coordinates based on the neighbors' weighted average (mean value edge weights).

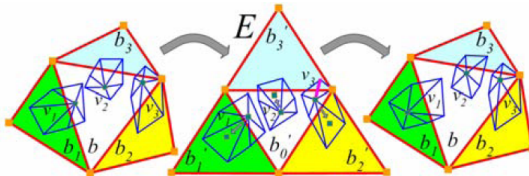
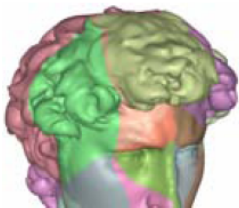


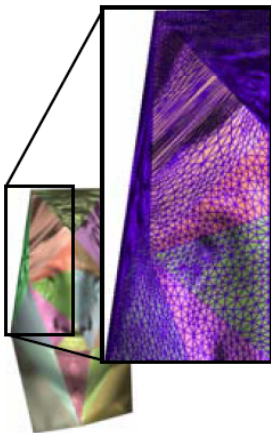
Figure 7: Relocating vertices on base mesh.



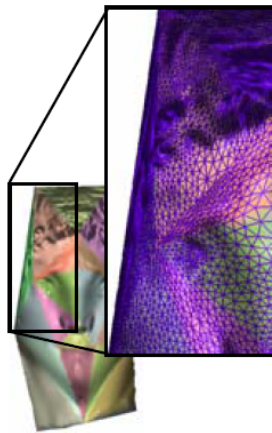
(a)



(b)



(c)



(d)



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- [2] O. Burghard, A. Dieckmann, and R. Klein. “Embedding shapes with Green’s functions for global shape matching”. In: *Computers & Graphics* 68 (2017), pp. 1–10.
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- [5] V. Kraevoy and A. Sheffer. “Cross-parameterization and compatible remeshing of 3D models”. In: *ACM Transactions on Graphics (TOG)* 23.3 (2004), pp. 861–869.
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